

Q1. Write a note on Cyber Security

Cyber Security refers to the practice of protecting computers, networks, servers, mobile devices, software, and data from unauthorized access, cyber attacks, damage, theft, or disruption.

The main purpose of cyber security is to ensure that information remains **confidential, accurate, and available** to authorized users.

Importance of Cyber Security

1. **Protects personal data** such as passwords, bank details, and identity information.
2. **Prevents unauthorized access** to computers and networks.
3. **Protects organizations** from data breaches and financial losses.
4. **Prevents malware and virus attacks.**
5. **Maintains privacy** of users and organizations.
6. **Ensures business continuity** by protecting systems from disruption.

Common Cyber Security Measures

- Using strong passwords and multi-factor authentication.
- Installing antivirus and anti-malware software.
- Keeping operating systems and applications updated.
- Using firewalls.
- Encrypting sensitive information.
- Regularly backing up important data.
- Educating users about phishing and other cyber threats.

Conclusion: Cyber security is essential in today's digital world because it protects information, systems, and users from continuously evolving cyber threats.

Q2. Explain CIA Triad of Information Security

The **CIA Triad** is a fundamental model of information security. CIA stands for:

C – Confidentiality

I – Integrity

A – Availability

1. Confidentiality

Confidentiality means ensuring that information is accessible **only to authorized users**.

Example: A bank account password should only be accessible to the account holder.

Methods:

- Passwords
- Encryption
- Access control
- Multi-factor authentication

2. Integrity

Integrity means ensuring that information remains **accurate, complete, and unaltered** by unauthorized people.

Example: A student's marks stored in a college database should not be changed without authorization.

Methods:

- Hashing
- Digital signatures
- Access controls
- Data validation

3. Availability

Availability means ensuring that information and systems are **accessible to authorized users whenever required**.

Example: An online banking website should be available when customers need to access their accounts.

Methods:

- Regular backups
- Disaster recovery
- Redundant systems
- Protection against DDoS attacks

CIA Triad Example

Principle	Meaning	Example
Confidentiality	Prevent unauthorized access	Password protection
Integrity	Prevent unauthorized modification	Hashing

Availability

Keep systems accessible

Backup servers

Conclusion: The CIA Triad provides the basic foundation for designing and maintaining secure information systems.

Q3. What are the different types of Cyber Crimes?

Cyber crime is any illegal activity in which a computer, network, digital device, or the internet is used as a tool or target.

Major Types of Cyber Crimes

1. **Hacking**
Unauthorized access to a computer system or network.
2. **Phishing**
Attackers send fake emails, messages, or websites to trick users into revealing passwords, banking information, or other sensitive data.
3. **Identity Theft**
Stealing someone's personal information and using it for fraudulent activities.
4. **Online Banking and Financial Fraud**
Unauthorized transactions or theft of money through online banking, cards, or digital payment systems.
5. **Cyber Stalking**
Repeatedly monitoring, threatening, or harassing a person using digital platforms.
6. **Cyber Bullying**
Using social media, messaging platforms, or other online services to threaten, insult, or harass someone.
7. **Ransomware Attacks**
Malware encrypts a victim's files and demands money in exchange for restoring access.
8. **Data Theft**
Unauthorized stealing or copying of confidential information from individuals or organizations.
9. **Denial-of-Service (DoS/DDoS) Attacks**
Overloading a server or network with traffic so that legitimate users cannot access the service.
10. **Email Spoofing**
Creating a fake email identity or sender address to deceive the recipient.
11. **Cyber Extortion**
Threatening a person or organization with releasing or damaging information unless a demand is fulfilled.

12. Software Piracy

Illegal copying, distribution, or use of copyrighted software.

Conclusion: Cyber crimes can cause financial losses, privacy violations, reputational damage, and disruption of services. Awareness and proper security practices are important to prevent them.

Q4. What are the different types of Malware?

Malware stands for **Malicious Software**. It is software intentionally designed to damage systems, steal information, gain unauthorized access, or disrupt computer operations.

Different Types of Malware

1. **Virus**
A virus attaches itself to a legitimate file or program and spreads when the infected file is executed.
2. **Worm**
A worm can replicate itself and spread automatically across networks without requiring a user to execute an infected file.
3. **Trojan Horse**
A Trojan disguises itself as legitimate or useful software but performs malicious activities after installation.
4. **Ransomware**
Ransomware encrypts files or locks systems and demands payment from the victim.
5. **Spyware**
Spyware secretly monitors user activities and may collect sensitive information such as passwords and browsing data.
6. **Adware**
Adware displays unwanted advertisements and may sometimes track user activity.
7. **Rootkit**
A rootkit is designed to hide malicious software or unauthorized activities deep inside a system.
8. **Keylogger**
A keylogger records keystrokes made by a user and can be used to steal passwords and other sensitive information.
9. **Bot/Botnet Malware**
Malware can turn infected devices into remotely controlled **bots**. A collection of such devices is called a **botnet**, which can be used for activities such as DDoS attacks.
10. **Fileless Malware**
Fileless malware operates primarily through legitimate system tools and memory

rather than relying on a traditional malicious executable file, making detection more difficult.

Difference between Common Malware

Malware	Main Characteristic
Virus	Attaches to files and spreads when executed
Worm	Self-replicates across networks
Trojan	Pretends to be legitimate software
Ransomware	Encrypts/locks data and demands payment
Spyware	Secretly monitors and collects information
Adware	Displays unwanted advertisements
Rootkit	Hides malicious activity
Keylogger	Records keystrokes
Botnet malware	Creates remotely controlled infected devices
Fileless malware	Operates mainly through memory/legitimate tools

Conclusion: Malware is one of the major threats to computer systems. Using updated software, antivirus protection, strong security practices, and avoiding suspicious links and files can help reduce malware infections.

Q1. Explain Cyberterrorism, its Meaning and Objective, and How It Is Different from Cyber Crime

Meaning of Cyberterrorism

Cyberterrorism refers to the use of computers, networks, or digital technologies to carry out attacks intended to cause serious disruption, fear, or damage to a government, society, or critical infrastructure for **political, ideological, or similar objectives**.

Cyberterrorism can target important digital systems such as government networks, electricity systems, transportation, communication systems, and financial infrastructure.

Objectives of Cyberterrorism

The major objectives include:

1. **Creating fear and panic** among the public.

2. **Disrupting critical infrastructure** such as electricity, transportation, and communication.
3. **Targeting government systems** and public services.
4. **Causing economic damage** to a country or organization.
5. **Supporting political or ideological objectives.**
6. **Stealing or disrupting sensitive information.**
7. **Reducing public confidence** in government and important institutions.

Cyberterrorism vs Cyber Crime

Cyberterrorism	Cyber Crime
Usually motivated by political, ideological, or similar objectives.	Usually motivated by financial gain, personal benefit, or other criminal purposes.
May target governments, critical infrastructure, or public systems.	Commonly targets individuals, businesses, or financial systems.
Attempts may be intended to create fear or large-scale disruption.	Often focuses on theft, fraud, unauthorized access, or data abuse.
Can have national-security implications.	Generally treated as a criminal and cybersecurity issue.
Example: An ideologically motivated attack intended to disrupt critical public infrastructure.	Example: Stealing someone's banking credentials through phishing.

Conclusion: Cyber crime is a broad category of illegal activity involving digital systems, whereas cyberterrorism generally involves cyber attacks connected to political or ideological objectives and intended to create serious disruption or fear.

Q2. What Are the Common Targets of Cyber Attacks? Explain with Examples and Their Impact on National Security and the Economy

Cyber attackers can target individuals, organizations, governments, and critical infrastructure.

Common Targets of Cyber Attacks

1. Government Systems

Government websites, databases, and internal networks can be targeted to steal sensitive information or disrupt public services.

Example: An attacker may compromise a government database containing confidential information.

2. Critical Infrastructure

Critical infrastructure includes systems such as:

- Electricity and power grids
- Water supply systems
- Transportation
- Telecommunications
- Healthcare
- Energy systems

Example: An attack on an electricity-management system could disrupt power distribution.

3. Financial Institutions

Banks, stock exchanges, payment systems, and financial companies are attractive targets because they contain valuable financial information.

Example: Attackers may use malware or stolen credentials to conduct unauthorized transactions.

4. Healthcare Organizations

Hospitals and healthcare systems store sensitive personal information and depend heavily on computer systems.

Example: A ransomware attack could prevent hospital staff from accessing important computer systems.

5. Businesses and Corporations

Organizations can be targeted for financial gain, intellectual-property theft, or industrial espionage.

Example: Attackers may steal confidential business documents or customer databases.

6. Military and Defence Systems

Defence organizations may be targeted to obtain classified information or disrupt communication and operational systems.

7. Communication Networks

Internet service providers, telecommunications companies, and communication infrastructure can be targeted to disrupt communication.

Impact on National Security

Cyber attacks can:

- Disrupt essential government services.
- Expose sensitive government information.
- Affect defence and military communications.
- Damage critical infrastructure.
- Create public fear and uncertainty.
- Reduce trust in government institutions.
- Affect a country's ability to respond to emergencies.

Impact on the Economy

Cyber attacks can cause:

1. **Financial losses** due to theft and fraud.
2. **Business interruption** when systems become unavailable.
3. **Recovery costs** for restoring affected systems.
4. **Loss of customer confidence.**
5. **Loss of intellectual property and trade secrets.**
6. **Damage to an organization's reputation.**
7. **Reduced productivity** during and after an attack.

Conclusion: Cyber attacks can affect both national security and economic stability because modern governments and businesses depend heavily on interconnected digital systems.

Q3. What Measures Can Governments and Organisations Take to Prevent and Respond to Cyberterrorism? What Is the Role of Cyber Security?

Cyberterrorism requires cooperation between **governments, organizations, cybersecurity professionals, and individuals.**

Measures Governments Can Take

1. Develop Cybersecurity Policies

Governments should establish national cybersecurity policies, standards, and legal frameworks for protecting critical infrastructure.

2. Protect Critical Infrastructure

Important systems such as power, transportation, healthcare, and communication networks should have strong security controls.

3. Establish Incident Response Teams

Governments should maintain specialized teams that can detect, investigate, and respond to major cyber incidents.

4. Information Sharing

Government agencies should share information about emerging threats, vulnerabilities, and attacks with relevant organizations.

5. International Cooperation

Cyber attacks can originate from different countries. Governments can cooperate internationally to investigate threats and strengthen cybersecurity.

6. Cybersecurity Training

Government employees and critical-infrastructure workers should receive regular cybersecurity awareness and incident-response training.

Measures Organisations Can Take

1. Strong Access Control

Only authorized employees should be allowed to access sensitive systems.

2. Multi-Factor Authentication

MFA provides an additional security layer beyond passwords.

3. Regular Software Updates

Operating systems, applications, and security tools should be regularly patched to reduce known vulnerabilities.

4. Network Security

Organizations should use appropriate firewalls, network segmentation, intrusion detection, and monitoring systems.

5. Data Backup

Important data should be backed up regularly and protected from unauthorized access.

6. Employee Awareness

Employees should be trained to identify phishing emails, suspicious links, malware, and social-engineering attacks.

7. Incident Response Plan

Organizations should have a documented plan explaining how to detect, contain, investigate, recover from, and learn from a cyber incident.

Role of Cyber Security in Combating Cyberterrorism

Cybersecurity plays a major role by providing **prevention, detection, response, and recovery** capabilities.

Prevention

Security controls such as firewalls, encryption, MFA, secure configurations, and regular patching help prevent unauthorized access.

Detection

Security monitoring, intrusion detection systems, endpoint security, and log analysis can help identify suspicious activity.

Response

When an attack occurs, incident-response teams can isolate affected systems, investigate the incident, and limit its impact.

Recovery

Backups, disaster-recovery systems, and business-continuity plans help organizations restore essential services.

Awareness

Cybersecurity education helps employees and citizens recognize common threats such as phishing and social engineering.

Conclusion

Cybersecurity is an important component of national security because governments, businesses, and critical infrastructure increasingly depend on digital systems. **Strong security controls, continuous monitoring, trained personnel, effective incident response, information sharing, and international cooperation** can reduce the risk and impact of cyberterrorism.